DigitalWelfare: Project Documentation

A one-stop web platform to discover, check eligibility for, compare and claim government welfare schemes

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Theme: Governance and Civic Technology

Abstract

India's central and state governments run hundreds of welfare programs covering agriculture, education, health, women and child development, social justice, employment and more. Despite this, many eligible families never receive these benefits because scheme information is scattered across dozens of ministries and portals, eligibility rules are buried in lengthy official documents, citizens do not know which documents to carry, and application deadlines are missed.

DigitalWelfare is a Progressive Web Application that addresses these gaps. It provides a searchable directory of Central and Maharashtra government schemes, an instant eligibility matcher based on age, gender, income, occupation, social category and state, an estimated annual benefit calculator, a tailored document checklist, a side-by-side comparison of up to three schemes, alert subscriptions, and a printable two-page facilitation dossier for Common Service Centre (CSC) and Setu operators.

The platform is built with Next.js 16, React 19, TypeScript, Tailwind CSS v4, Firebase (Authentication, Cloud Firestore and Admin SDK) and Cheerio. A 10-stage automated data pipeline fetches scheme data from official portals, validates and cleans it, translates Marathi/Hindi text, extracts eligibility rules, documents and benefit values, and updates Firestore. Service workers and a web manifest make the app installable and usable offline, with a built-in fallback catalog for zero-downtime resilience.

The result is a free, citizen-friendly portal that reduces the time, travel and confusion involved in accessing welfare benefits.

Chapter 1: Introduction

1.1 Introduction to Government Welfare Schemes and Digital Access

Government welfare schemes are programs funded by central and state governments to support citizens in areas such as livelihood, education, health, housing, social security and employment. Examples include PM-KISAN for farmers, the Mukhyamantri Majhi Ladki Bahin Yojana for women, post-matric scholarships for students, and MJPJAY / PM-JAY for health cover of up to ₹5 lakh per year.

Under the Digital India initiative, most schemes now accept applications online through portals such as MahaDBT, MP-SIMS and MahaSwayam. However, digital availability does not automatically mean digital accessibility: citizens must first know that a scheme exists, understand whether they qualify, and prepare the right documents.

1.2 Background of the Problem in India

Central and state governments spend thousands of crores each year on welfare, yet millions of eligible families miss out on benefits. The main causes are:

- **Scattered information:** Schemes are spread across 50+ central ministries and numerous state portals, making discovery difficult.
- **Confusing eligibility rules:** Rules are buried in long official gazettes. Citizens struggle to tell whether an income cap is per person or per family, or whether their land holding or category qualifies.
- **Missing documents:** People travel long distances to a CSC or Setu office and wait in queues, only to be told that a document (for example a Tehsildar income certificate, 7/12 land record or caste validity certificate) is missing.
- **Missed deadlines:** By the time information reaches a village or community, the application window has often closed.

1.3 Problem Statement

There is no simple, unified, citizen-friendly system that lets an ordinary person discover relevant welfare schemes, understand in plain terms whether they are eligible, know exactly which documents to carry, and be alerted before deadlines. As a result, eligible citizens lose benefits, and facilitation centres spend time on avoidable rejections and repeat visits.

1.4 Need and Motivation

- Eligible families lose financial support they are entitled to.
- Rural and low-literacy users need a clear, simple and mobile-friendly interface.
- CSC / Setu operators need a quick way to verify eligibility and print a document checklist.
- Rural connectivity is poor, so the solution must work offline.
- Scheme data changes frequently, so manual maintenance is not sustainable; an automated update pipeline is needed.

1.5 Aim

The aim of DigitalWelfare (Aayush) is to give every citizen a simple, friendly, one-stop portal to search schemes, check eligibility with one click, calculate total financial benefits, and obtain an exact checklist of documents before visiting an application centre.

1.6 Objectives

1. Build a searchable, filterable directory of Central and Maharashtra government welfare schemes.
2. Provide an instant eligibility matcher based on a citizen's basic profile.
3. Estimate the annual financial benefit a citizen can receive.
4. Generate a tailored document checklist and a printable facilitation dossier.
5. Allow side-by-side comparison of up to three schemes.
6. Offer alert subscriptions via WhatsApp, SMS or Email for new schemes and approaching deadlines.
7. Automate scheme collection, cleaning, translation and rule extraction through a data pipeline.
8. Deliver an installable, offline-capable PWA suited to real-world rural conditions.
9. Protect data with role-based access, server-side admin operations and Firestore security rules.

1.7 Scope

In scope: Central and Maharashtra schemes across ten departmental groups (education, social justice, tribal development, OBC/VJNT/SBC welfare, agriculture, women and child welfare, youth and employment, public health, labour/construction workers, and divyang and senior welfare); eligibility matching; document checklists; comparison; alerts; admin hub; PWA and offline mode.

Out of scope (current version): Submitting applications on behalf of citizens, DigiLocker verification, other states' full catalogs, voice search and a WhatsApp bot (these are in the roadmap).

1.8 Target Users and Beneficiaries

User group	How they benefit
Citizens (farmers, students, women, seniors, workers, persons with disabilities)	Discover schemes, check eligibility, know required documents
CSC / Setu operators and field workers	Print a 2-page dossier, serve more people with fewer rejections
Administrators	Monitor pipeline updates, subscribers and scheme records
NGOs and community workers	Guide communities using a single trusted tool

1.9 Expected Social Impact

- Higher scheme uptake among eligible families.
- Fewer wasted trips and application rejections caused by missing documents.
- Greater awareness of deadlines through alerts.
- Reduced workload and queues at facilitation centres.
- Stronger digital inclusion in line with Digital India.

Chapter 2 : Existing System and Proposed Solution

2.1 How Citizens Currently Discover Schemes

Citizens currently rely on word of mouth, local officials, newspaper notices, social media forwards, CSC/Setu centres, and individual government portals (for example MahaDBT or ministry websites). Each portal covers only its own department, and the information is often in long PDFs or official notifications.

2.2 Problems with Scattered Information and Complicated Eligibility

- No single place lists schemes from all departments.
- Eligibility conditions (age, income, category, occupation, land size, domicile) are written in legal language.
- The same condition can be interpreted differently (per-person vs per-family income).
- Many portals are not optimized for mobile devices or regional languages.

2.3 Difficulties in Identifying Required Documents and Deadlines

- The document list differs by scheme and by applicant category (for example caste validity for reserved categories, 7/12 extract for farmers).
- Citizens learn about missing documents only at the centre.
- Deadlines are announced irregularly and are easy to miss.

2.4 Limitations of the Existing Approach

1. Manual, time-consuming search across many sources.
2. No personalized eligibility result.
3. No consolidated benefit estimate.
4. No tailored document checklist.
5. No comparison between similar schemes.
6. No proactive alerts.
7. Poor offline support and weak usability on low-end phones.

2.5 Overview of the Proposed DigitalWelfare Platform

DigitalWelfare (Aayush) is a unified PWA that aggregates scheme data through an automated pipeline, stores it in Cloud Firestore, and presents it through simple pages: scheme directory (/schemes), eligibility matcher (/eligibility-check), comparison (/compare), alert subscription (/api/subscribe-alerts), and an admin hub (/admin). A fallback catalog (lib/fallback-schemes.ts) ensures the app works even if the database or network is unavailable.

2.6 Comparison: Existing System vs Proposed System

Aspect	Existing system	DigitalWelfare (proposed)
Scheme discovery	Scattered across 50+ ministries and state portals	Single searchable directory with category filters
Eligibility check	Manual reading of gazettes	One-click matcher from basic profile
Benefit estimate	Not available	Estimated annual benefit per scheme and in total
Documents	Learned at the centre	Tailored checklist before visiting
Comparison	Not available	Up to 3 schemes side by side, printable
Alerts	Rare, informal	WhatsApp / SMS / Email subscriptions
Language	Mostly English; some Marathi/Hindi	Devanagari text normalized and translated in the pipeline
Data updates	Manual	Automated 10-stage pipeline
Connectivity	Needs stable internet	PWA with offline fallback
Privacy	Varies	No personal data needed for browsing, no tracking cookies

2.7 Advantages and Expected Improvements

- Saves time, travel and effort.
- Reduces application errors and rejections.
- Provides clarity through plain-language eligibility results.
- Works on any phone or desktop, even with poor connectivity.
- Keeps data fresh through automation.
- Helps facilitation centres with printable dossiers.

Chapter 3 : Requirements Analysis

3.1 Functional Requirements

ID	Requirement
FR1	The system shall allow citizens to register and log in (verify which login options are implemented).
FR2	The system shall let users search schemes by keyword.
FR3	The system shall filter schemes by category (Agriculture, Higher Education, Health, Women & Child, Social Justice, Skill Development, etc.).
FR4	The system shall show scheme details: benefits, eligibility criteria, required documents and the official portal link.
FR5	The system shall let users bookmark schemes.
FR6	The system shall accept age, gender, annual family income, occupation, social category and state and return all matching schemes.
FR7	The system shall calculate an estimated annual financial benefit.
FR8	The system shall generate a tailored document checklist.
FR9	The system shall generate a printable 2-page facilitation dossier.
FR10	The system shall compare up to three schemes and allow export / print.
FR11	The system shall let users subscribe to alerts via WhatsApp, SMS or Email.
FR12	The system shall provide a protected admin dashboard for pipeline sync, subscriber statistics and scheme management.
FR13	The system shall automatically fetch, clean, normalize and store scheme data.
FR14	The system shall work offline using cached assets and a fallback catalog.

3.2 Non-Functional Requirements

Category	Requirement
Performance	Fast loading using ISR, client-side caching and skeleton loaders with no layout shift
Usability	Simple, mobile-first, readable UI suitable for first-time users
Reliability	Fallback catalog ensures zero-downtime browsing
Availability	Offline access through service worker
Security	Role-based access, server-side admin writes, secrets in environment variables
Privacy	No personal data needed for browsing; no tracking cookies
Scalability	Cloud Firestore scales with load; composite indexes defined
Maintainability	Modular TypeScript code with a clear pipeline and component structure
Portability	Runs on modern browsers, installable on phones and desktops
Accuracy	Links to official portals so users can verify details

3.3 Hardware Requirements

Role	Minimum
Developer machine	Dual-core CPU, 8 GB RAM, 5 GB free disk, internet connection
Server / hosting	Node.js-capable cloud platform [name your host, e.g. Vercel]
End user	Any smartphone, tablet or PC with a modern browser and (optionally) internet

3.4 Software Requirements

- Node.js 18.x or 20.x and npm
- Next.js 16 (App Router), React 19, TypeScript 5, Tailwind CSS v4, Lucide React
- Firebase project with Authentication and Cloud Firestore; Firebase Admin SDK
- Cheerio for HTML parsing
- Git / GitHub, ESLint, a code editor such as VS Code
- Browsers: current Chrome, Edge, Firefox, Safari

3.5 User Roles

- Citizen (public user): Browse, search, compare, check eligibility, bookmark, subscribe to alerts.
- Facilitation operator (CSC / Setu): Uses the same public features and prints the dossier.
- Administrator: Runs pipeline sync, views subscriber stats, manages scheme records; access protected by an admin passcode and server-side privileges.

3.6 Use Cases

1. As a farmer, I want to enter my details and see which schemes I qualify for, so that I know my annual benefit (for example ₹12,000/year from PM-KISAN + Namo Shetkari).
2. As a woman applicant, I want to learn whether I can get the Ladki Bahin benefit of ₹1,500 per month.
3. As a student, I want to see the document list for a scholarship so that I do not return to the office.
4. As a CSC operator, I want to print a two-page dossier for each visitor.
5. As a citizen, I want to compare two or three schemes before choosing.
6. As a rural user, I want to open the app offline.
7. As an administrator, I want to trigger a pipeline sync and verify updates.

3.7 Assumptions and Constraints

- Scheme information on official portals is publicly accessible and legally usable.
- Government site structure may change, which can break parsers.
- Eligibility results are indicative; final approval rests with the issuing department.
- Current coverage focuses on Central and Maharashtra schemes.
- Firebase free-tier quotas apply.

3.8 External Services and Dependencies

Service	Use
Firebase Authentication	Citizen login
Cloud Firestore	Scheme, subscriber and user data
Firebase Admin SDK	Privileged server-side writes
Government portals (MahaDBT, MP-SIMS, MahaSwayam, central portals)	Source data and application redirection
Messaging channel for alerts	WhatsApp / SMS / Email delivery
Hosting platform	Deployment of Next.js app

Chapter 4: System Design and Architecture

A. System Architecture

Purpose: components and hosting.

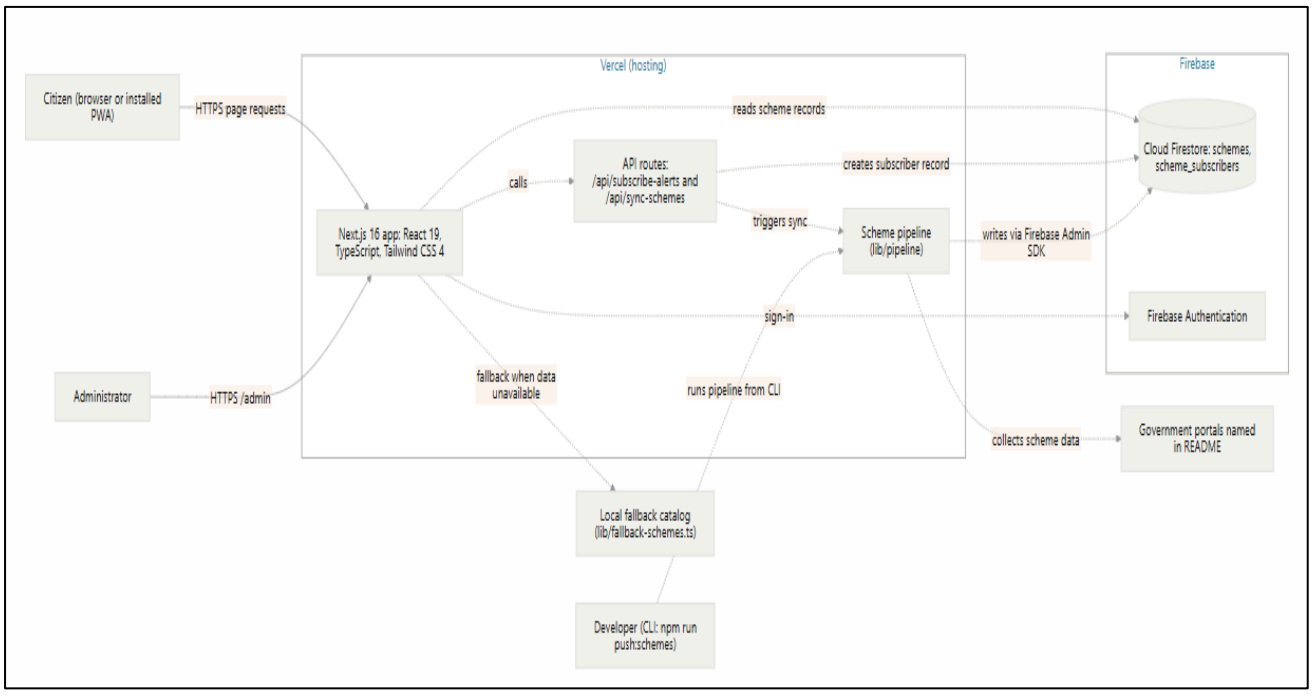


Fig 4.1

B. Data Flow Diagrams

Notation: rectangle = external entity, circle = process, cylinder = data store. Mermaid has no native DFD shapes, so these are approximations.

B1. Context-level DFD

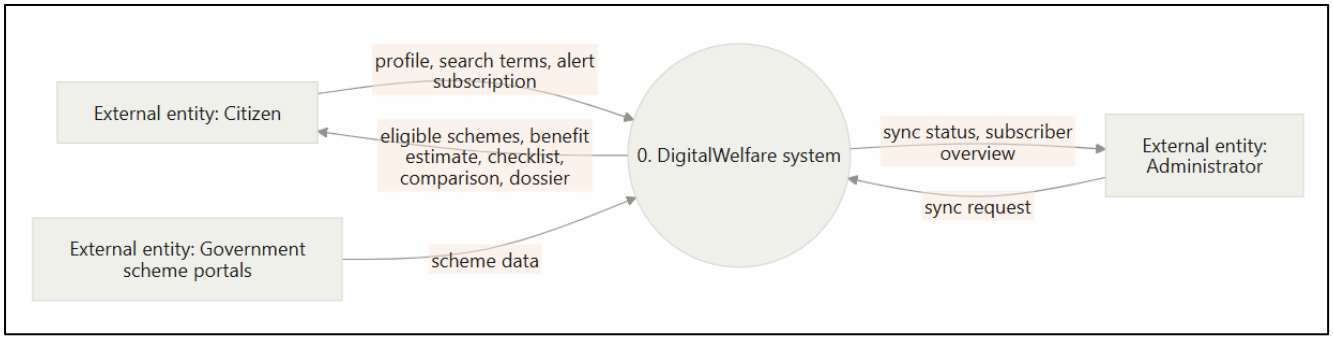


Fig 4.2

B2. Level 1 DFD

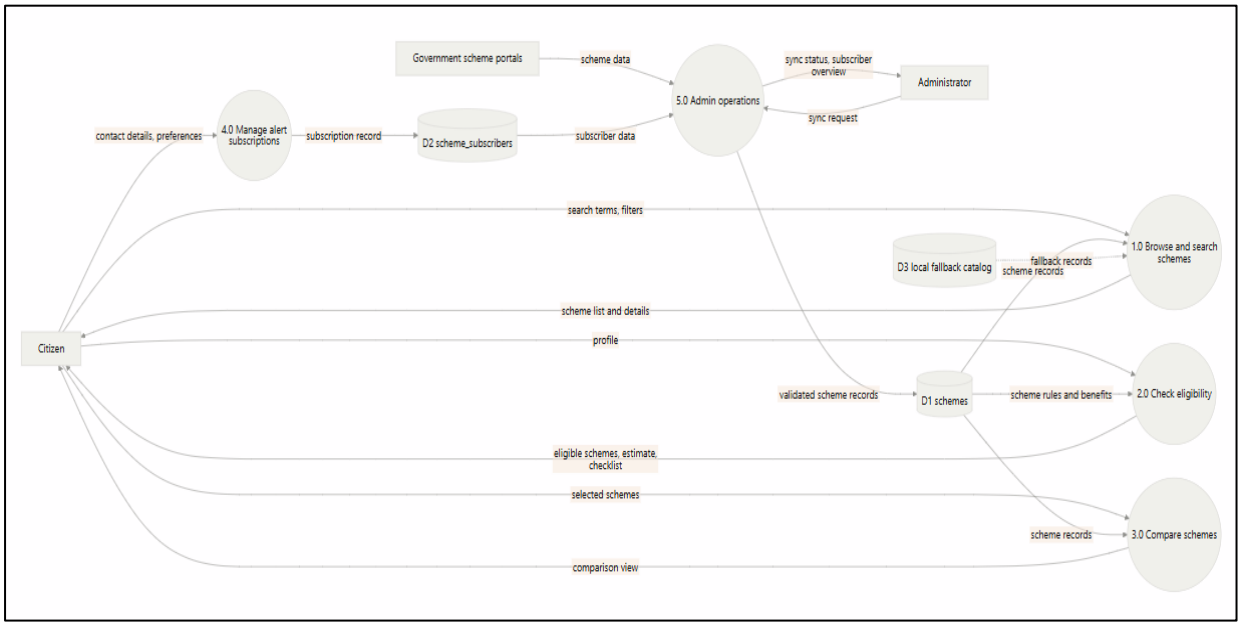


Fig 4.3

C. Application Workflow

1. Citizen Opens the Portal: The citizen visits the portal and chooses one of the three main features: Eligibility Check, Browse Schemes, or Compare Schemes.

2. Browse Government Schemes: The citizen can access the schemes directory, search schemes using keywords, or filter them by category to find relevant government schemes.

3. View Scheme Details: After selecting a scheme, the citizen can view its details, eligibility conditions, benefits, and other important information. They can also optionally subscribe to alerts for updates.

4. Check Eligibility: The citizen enters personal details such as age, gender, state/UT, income, occupation, and social category.

5. Validate Details and Calculate Eligibility: The system checks whether all required fields are filled. If any information is missing, the citizen is asked to complete it. Otherwise, the system calculates eligibility.

6. View Eligible Benefits: The portal displays eligible schemes, estimated benefits, and a checklist of required documents. The citizen can also optionally print a facilitation dossier.

7. Compare and Apply: Citizens can select up to three schemes for a side-by-side comparison. After reviewing the details, they can proceed to the respective official

government portal to complete the application. The actual application process takes place outside this system.

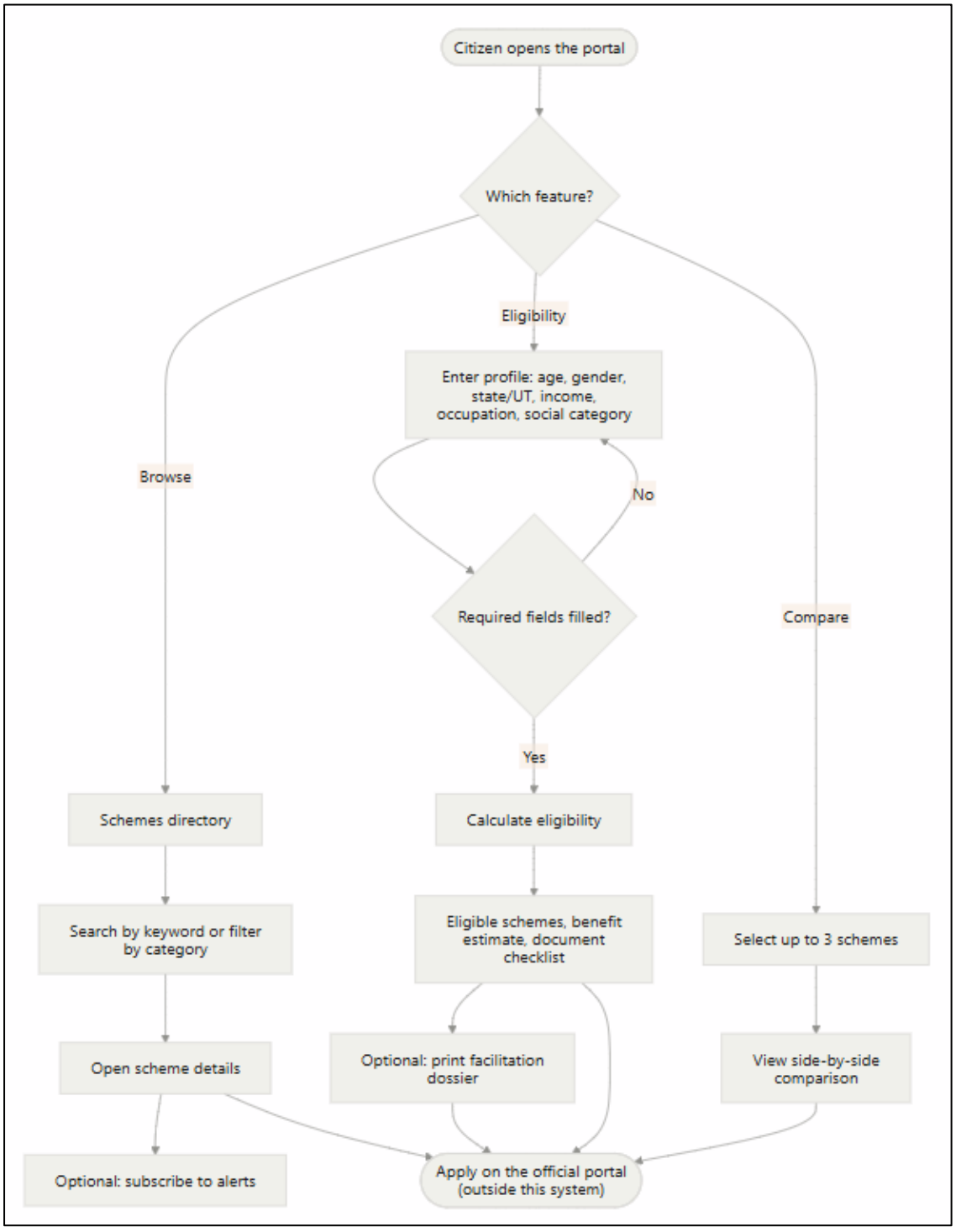


Fig 4.4

D. Use Case Diagram

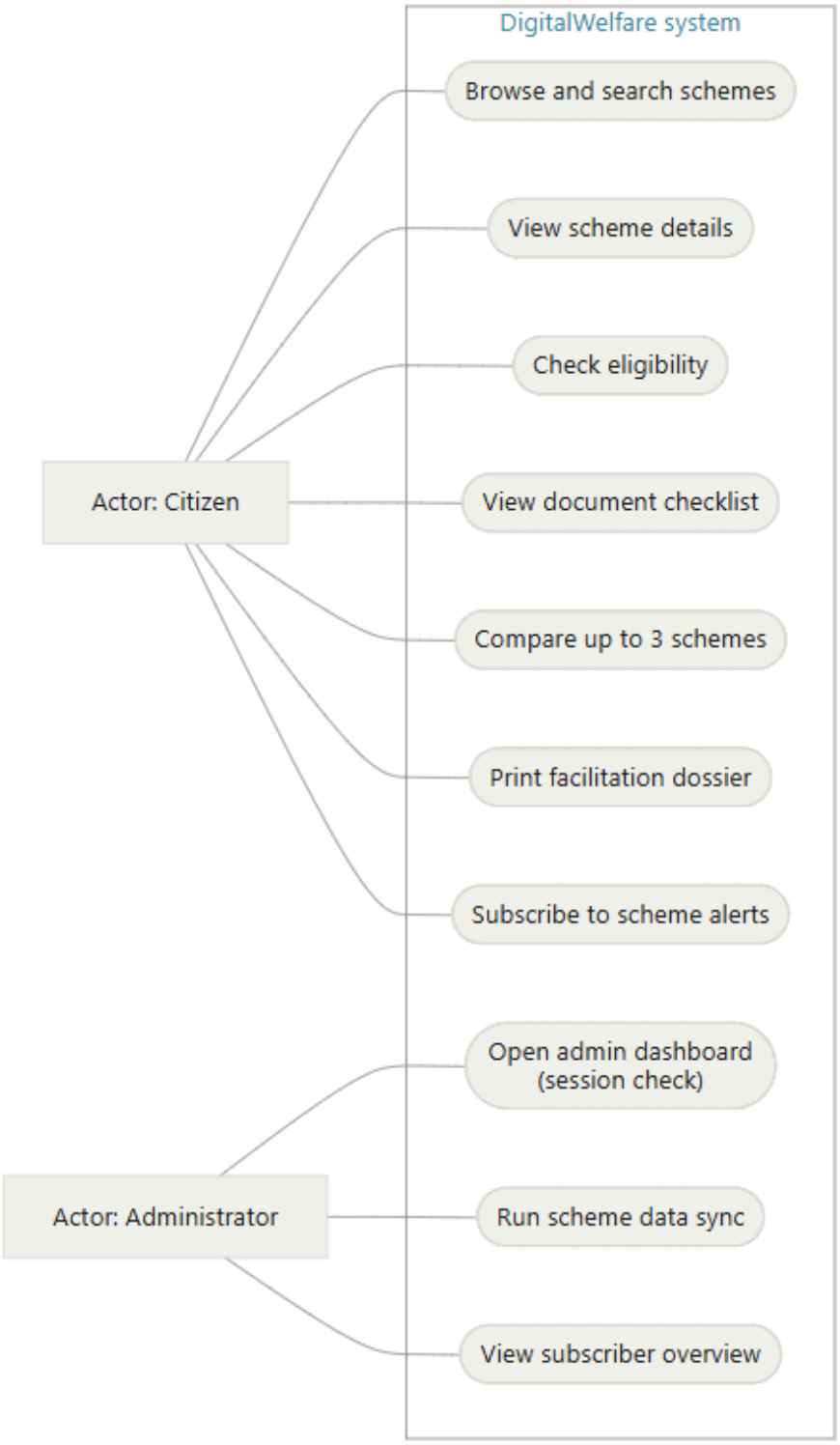


Fig 4.5

E. Sequence Diagrams

E1. Alert subscription

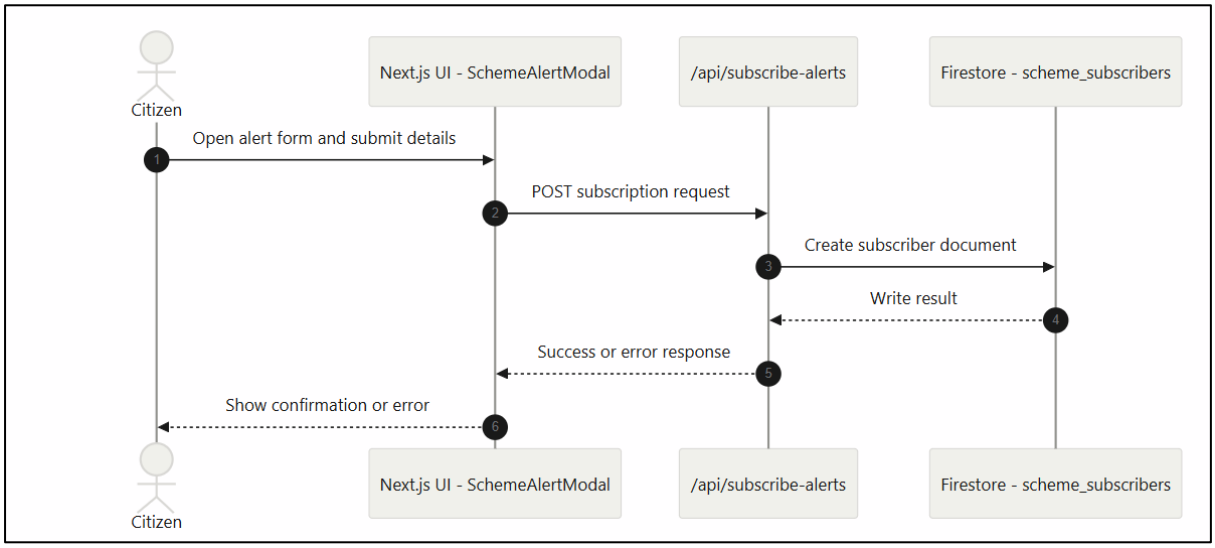


Fig 4.6

E2. Admin data sync

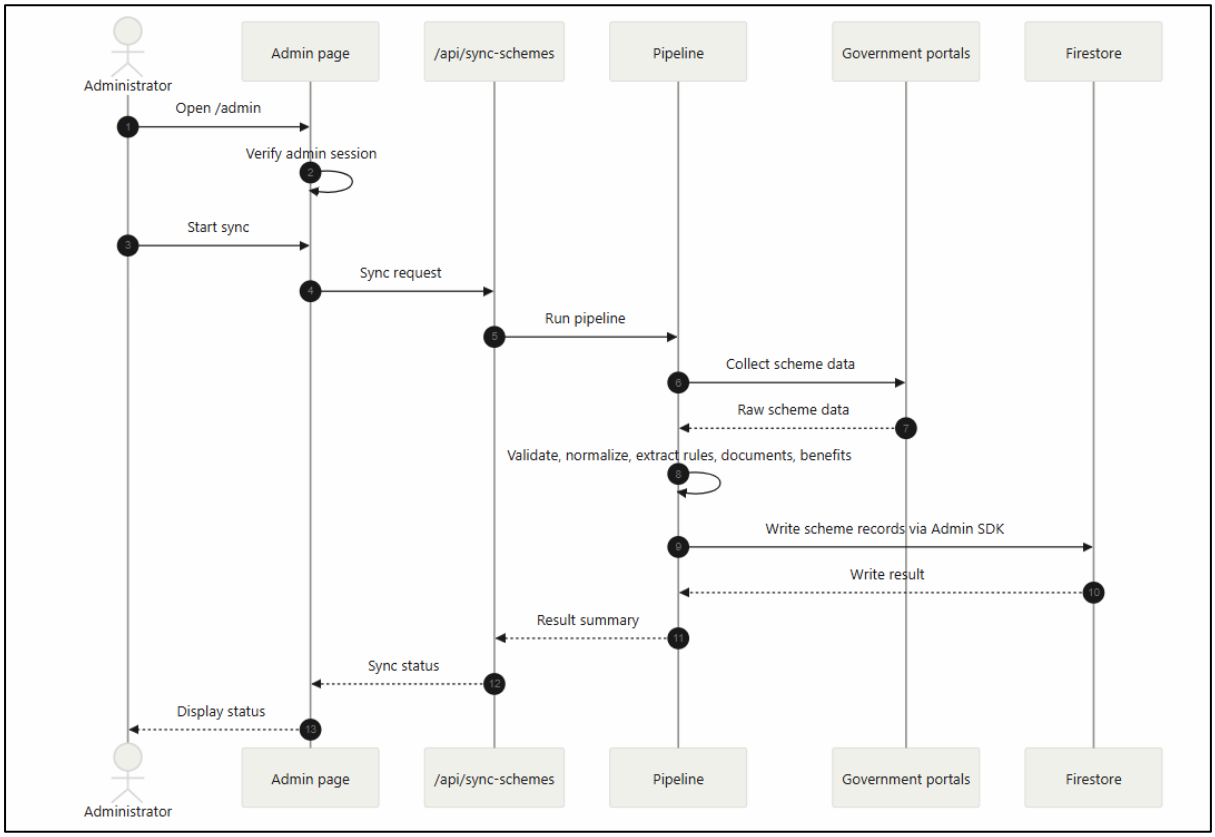


Fig 4.7

G. Scheme Data Pipeline

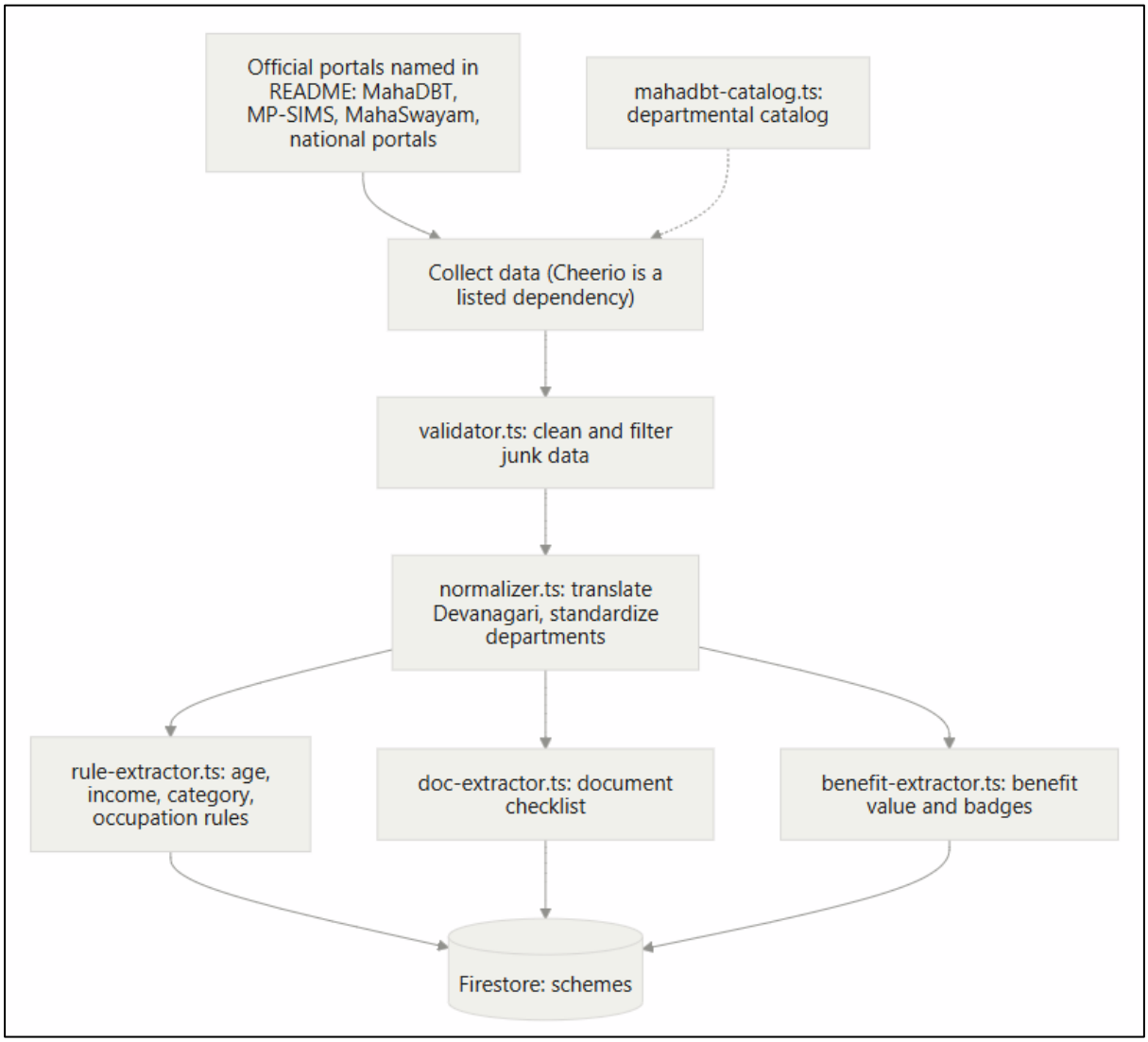
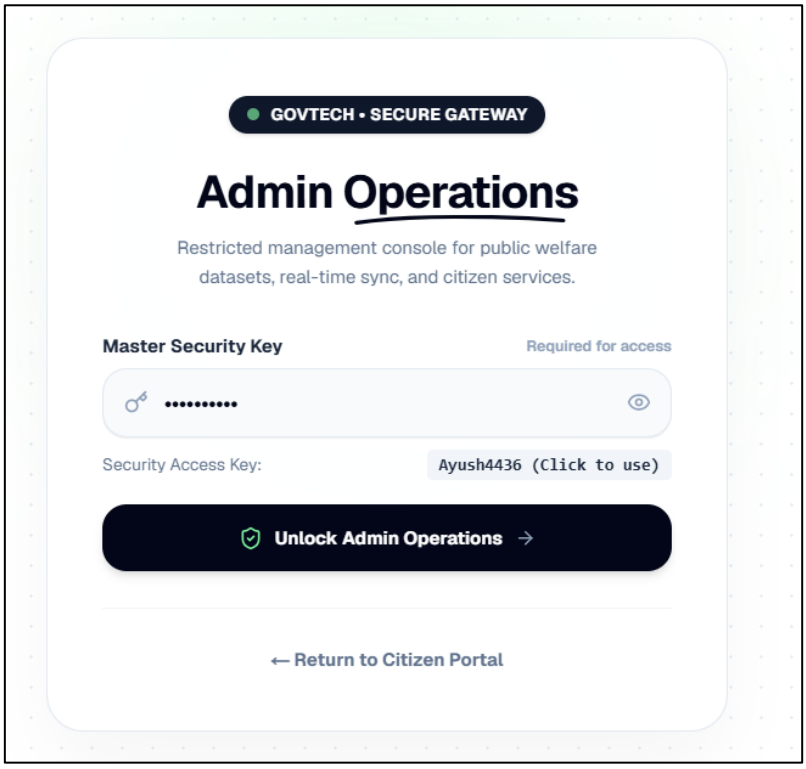
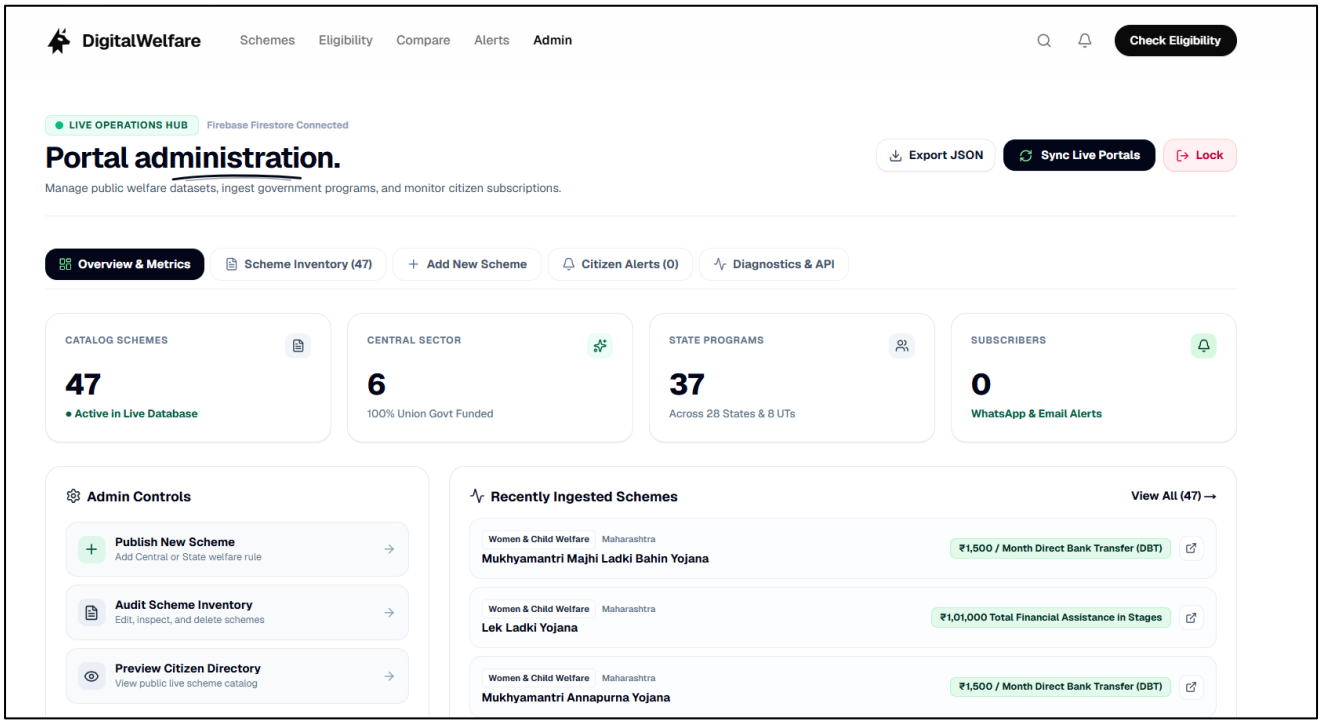


Fig 4.8

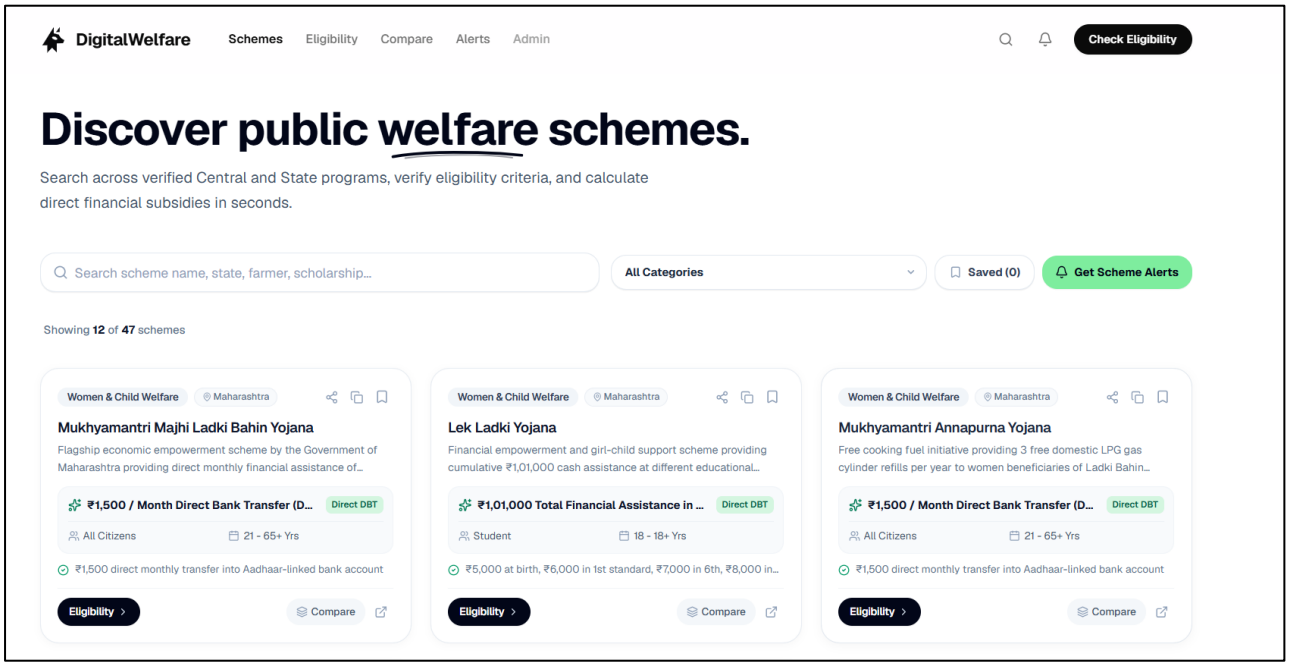
Chapter 5: Project Screenshots

1] Admin Page

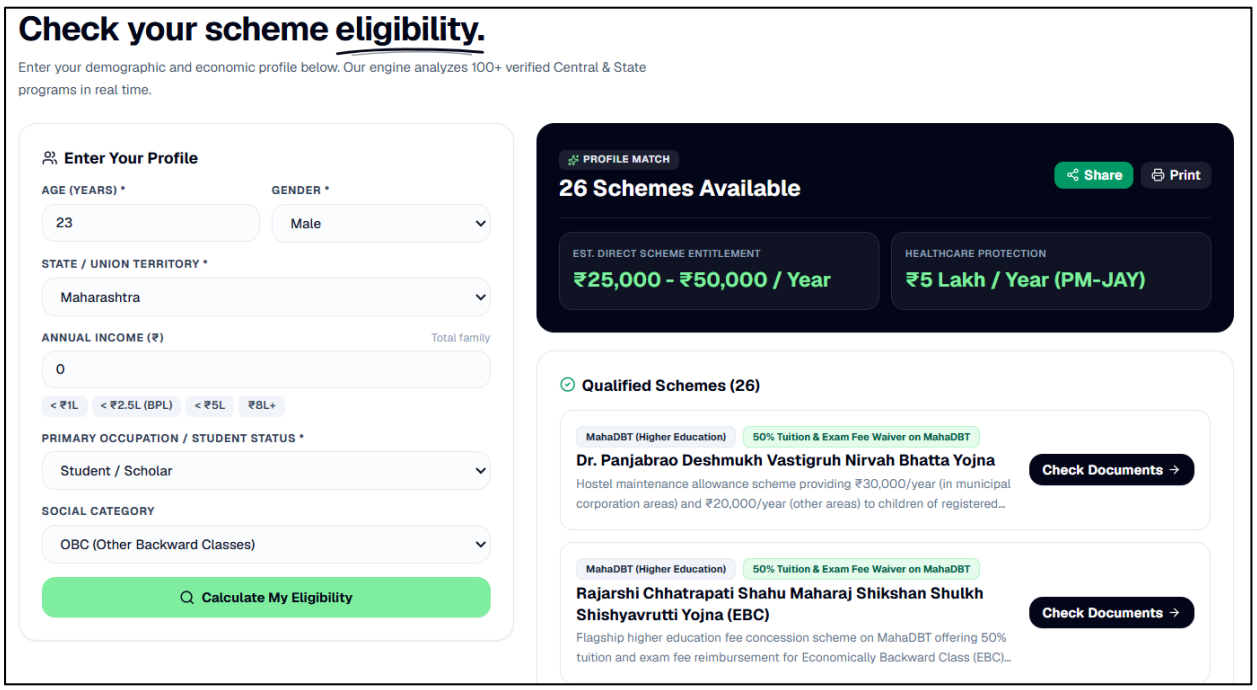




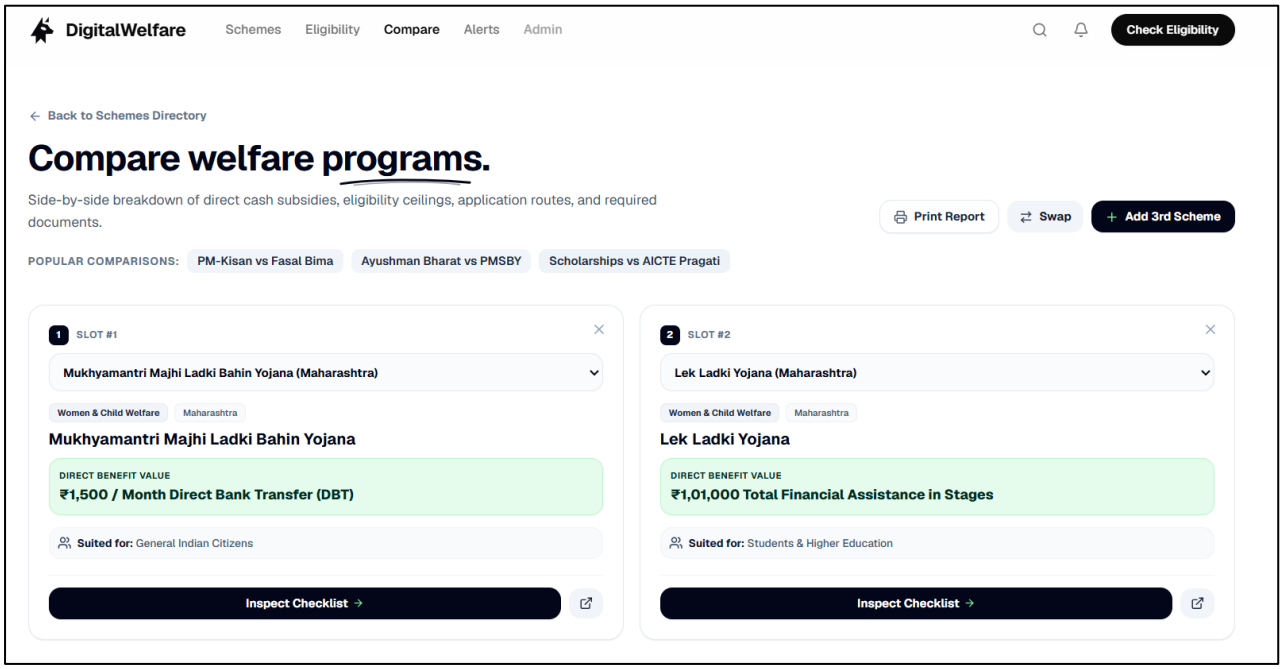
2] Home/Dashboard Page



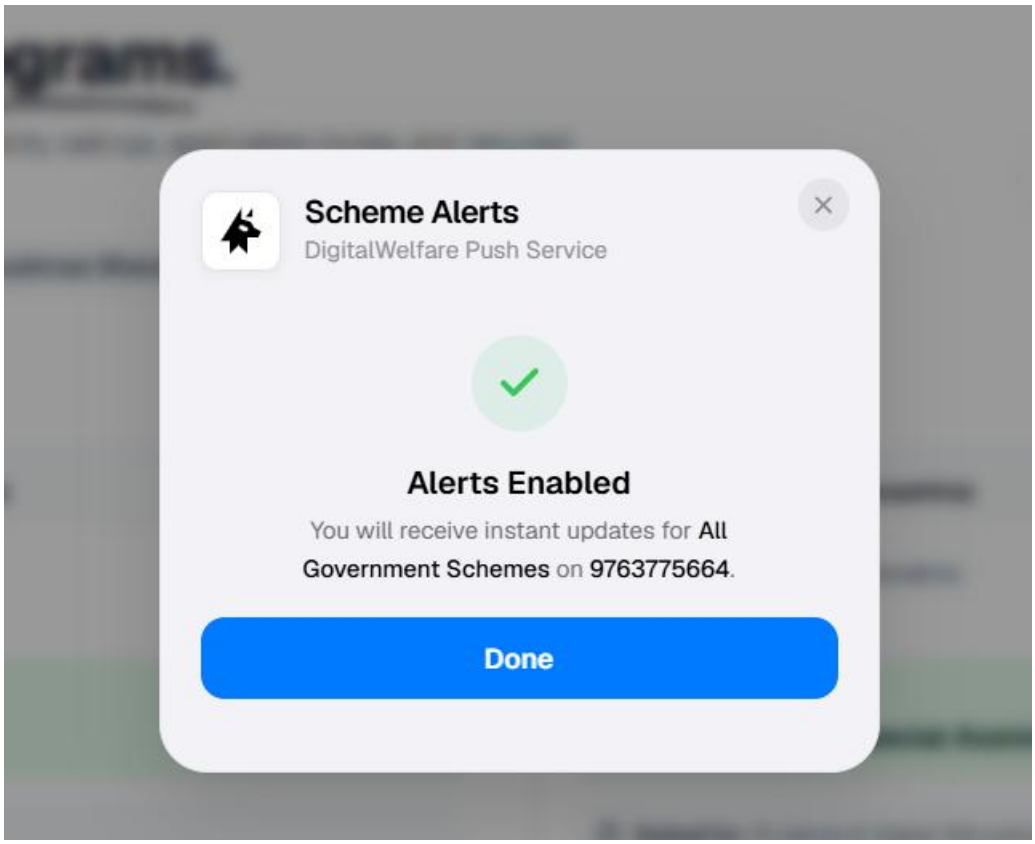
3] Eligibility Checker Page



4] Schemes Comparison Tab



5] Alerts for Schemes



Chapter 6: Future Scope and Conclusion

Future Scope:

- **Voice Search in Regional Languages:** Speak to search schemes in Hindi, Marathi, Tamil, or Telugu using speech recognition.
- **User Login:** A citizen login and profile management module can be added in future versions to provide personalized user access
- **DigiLocker Integration:** One-click verification of Aadhaar, caste certificates, and marksheets.
- **AI Application Assistant:** Step-by-step interactive helper for filling out complex government forms**.**
- **WhatsApp Bot:** Check eligibility and receive scheme alerts directly on WhatsApp.

Conclusion:

The **DigitalWelfare** project aims to simplify the process of discovering and accessing government welfare schemes through a centralized, user-friendly digital platform. It addresses common challenges faced by citizens, such as scattered scheme information, confusing eligibility criteria, missing documentation, and lack of awareness about application deadlines.

The system provides features such as scheme browsing and searching, automated eligibility checking, estimated benefit calculation, personalized document checklists, side-by-side scheme comparison, and scheme update alerts. It also uses web scraping and an automated data pipeline to collect, validate, organize, and update scheme information from government portals. Additionally, user authentication, administrative controls, and Progressive Web App (PWA) support help improve accessibility and management.

Overall, DigitalWelfare acts as a bridge between citizens and government welfare initiatives by making relevant information easier to find, understand, and access. It aims to reduce information gaps, save citizens' time and effort, and encourage more people to take advantage of the benefits available to them. The platform directs users to the respective official government portals to complete their applications, ensuring that the actual application process remains with the authorized government systems.